

Supporting Information for “Evaluation of short-term probabilistic eruption forecasting at Whakaari, New Zealand.”

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Introduction

Supplementary information includes additional description and figures that provide further background for the models presented in the main text.

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Text S1.

Systematic Time-series feature engineering for eruption forecasting.

Here, we provide further description of the Whakaari eruption forecast model presented in Dempsey et al. (2020).

Data. We use various measures of tremor as raw data that are informative of the volcano state. RSAM (real-time seismic amplitude measure) is a 10-minute moving average of vertical station velocity, which is bandpass filtered between 2 and 5 Hz. MF (medium frequency) and HF (high frequency) are computed the same way as RSAM but with bandpass filters of 4.5 to 8, and 8 to 16 Hz. DSAR (displacement seismic amplitude ratio) is the ratio of the integrals of MF and HF signals (Caudron et al., 2019). Transforms of the data – differencing, logarithm and inverse – were tested in Dempsey, Cronin, Mei, and Kempa-Liehr (2020) but were ultimately not found useful for forecasting. In this study, we used data from 1 Jan 2011 to 1 Jul 2021.

Windows. The raw tremor time series are subdivided into windows of 48 hours in duration (each comprising 288 10-minute samples). At low resolution, adjacent windows overlap each other by 36 hours, yielding more than 7600 windows. From each window, we extract more than 700 time series features using the Python package tsfresh (Christ et al., 2018). These include measures of distribution (mean, standard deviation, number of peaks), correlation (lags, Fourier coefficients), linearity (slope and intercept of linear regressors) and other information quantities (entropy, energy, nonlinearity scores). All feature values are contained in a matrix, whose individual columns represent the different features, and whose rows characterize a specific window. The rows are indexed by the timestamp of the last

measurement in the window. Other window lengths were tested in Dempsey et al. (2020) and found to be not as effective.

Binary labels. We define a classification problem in which we are interested in signals occurring in the 48 hours prior to eruptions. All such windows are labeled 1 (five eruptive periods, 20 labels of ‘1’ in total) and the remaining are labeled 0. Note, the two subsequent eruptions in the days following the 3 Oct 2013 eruption are not labeled ‘1’ to avoid biasing the model to this particular eruptive period.

Feature significance. Important features are assessed on whether windows with label ‘1’ have significantly different values to those with label ‘0’. Significance is assessed using a Mann-Whitney U test for the difference in distribution medians, and p-values are retained as significant providing they fall below a Benjamini-Yekutieli adjusted threshold, which controls the false discovery at 5% when performing large numbers of statistical tests.

Model training. Our classification model is a random forest comprising individual decision trees trained on different parts of the data. First, data for a test eruption is set aside and not used for training (Fig. S1). Then, non-eruptive, label ‘0’ windows are randomly discarded to achieve balance with the remaining 16 label ‘1’ windows. Features are ranked by statistical significance in this down-sampled dataset and the top 20 are retained. A decision tree with maximum depth 3 is then trained using a grid-search over hyperparameters and shuffle-split cross-validation. Hyperparameter search space is the same as reported in the supplementary information for Dempsey et al. (2020). This procedure is repeated 500 times for different undersamplings and the average of all so-trained decision trees becomes the model output. Other classification models were tested in Dempsey et al. (2020) and found to be not as effective. The random number generator was reseeded with five consecutive indices and this did not affect the model results.

Model forecasting. For the test period of interest, 48-hour data windows are computed with each subsequent window advanced by 10-minutes. Time series features are computed for these windows and evaluated by each trained decision tree in the random forest. The binary outputs are averaged to produce a value between 0 and 1. The output is compared against a preset threshold and an alarm is issued if it exceeds this value. An alarm is only rescinded after 48 consecutive hours of model output below the threshold. Alternatively, the model output can be passed to a second function that returns a calibrated probability output.

Model output uncertainty. The random forest structure reports the average output of N decision trees. Each tree is trained on a random subset of the data and issues a binary prediction (0 or 1) for a future eruption. Seeding the random number generator differently would produce a different set of decision trees that, when shown the same data, would return a different average outcome from the random forest.

This uncertainty can be estimated by constructing a binomial proportion confidence interval. We conceptualise construction and evaluation of each decision tree as a Bernoulli trial whose success-failure (eruption-no eruption) outcome is determined with probability, $\hat{p}$. For suitably large N , from the central limit theorem we write a confidence interval for $\hat{p}$ as

$$\hat{p} \pm z \sqrt{\frac{\bar{x}(1 - \bar{x})}{N}} \quad (1)$$

where $\bar{x}$ is the estimate of $\hat{p}$ obtained from the N decision trees and z is a quantile of the standard normal distribution (e.g., for a 95% confidence interval, $z = 1.96$.)

We tested the validity of Eq. (1) by simulating the confidence intervals numerically. First, we trained a population of $N_0=1000$ decision trees. Then, we constructed M random forests by sampling (with replacement) N trees from the population (where $N < N_0$). Finally, we verified that output variability of the so-constructed random forests was consistent with (1) across both N and $\bar{x}$ (Fig. S2).

Text S2.

Out-of-sample model performance.

Model performance with earthquake filtering and data normalization for all eruptive periods is shown in Fig. S3. To summarize:

1. The 2012 eruption is anticipated with high confidence. High model output is recorded for several days preceding the eruption. Instantaneous probability spikes above 20% about five hours prior, and time-averaged probability reaches about 4%.
2. Forecasting of the Aug 2013 eruption is less convincing, but still anticipated by an alarm triggered in response to a strong RSAM spike 30 hours before the eruption. Instantaneous probability peaks at about 5%.
3. The 3 Oct 2013 eruption is anticipated about four hours prior. Instantaneous and time-averaged probabilities reach around 3%. Retraining of the model on the first eruption in the sequence of three is necessary if the middle eruption is to be anticipated.
4. The 2016 eruption is missed. Probabilities remain low prior to it, and are not especially high afterward.
5. The 2019 eruption is anticipated with high confidence. An alarm is triggered by a precursor about 16 hours before the eruption. Instantaneous probability spikes briefly to 13%.

Text S3.

Stationarity of RSAM at Whakaari

Kilgour et al. (2021) note that the GeoNet operated station WIZ was upgraded in 2014. Although a precise date is unknown, we can check if any biases were introduced as a result of the upgrade. Fig. S4 plots histograms of RSAM data for four 3-year periods, spanning 2008-2011, 2011-2014, 2015-2018, and 2018-2021. Data from 2014 is excluded in this brief analysis. The figure shows that there are no obvious offsets in the distributions when comparing the pre-2014 and post-2015 periods. Therefore, there is no reason to believe that a station upgrade has introduced a major bias impacting this study.

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Table S1. Eruption elicitation probabilities and running calculations, from GeoNet (2020).

Date	24-hr eruption probability
2 Dec 2019	0.30-0.54% ^a
11 Dec	40-60%
12 Dec	50-60%
13 Dec	50-60%
14 Dec	35-50%
15 Dec	30-45%
16 Dec	30-40%
17 Dec	25-40%
18 Dec	20-40%
19 Dec	15-30%
20 Dec	10-20% ^b
23 Dec	2-10% ^c
6 Jan 2020	1-2% ^d

^aConverted from an 8 - 14% probability over a 4-week period

^bConverted from a 27 - 53% probability over a 72-hour period

^cConverted from a 25 - 77% probability over a 2-week period

^dConverted from a 23 - 50% probability over a 4-week period

Table S2. Summary of local forecast skill of data-driven models in the training period. FS_0 = Forecast Skill of uninformed model in 16-day period. $\widetilde{FS}$ = Scaled Forecast Skill of alarm-based and calibrated probability models. Local forecast skill shown for 16-day periods centred on each of six eruptions in the training set. Note, due to its manner of calculation, $FS_0 \approx -0.7 \times$ number of eruptions in the 16-day period.

27 Jul to 12 Aug 2012			11 to 27 Aug 2013			25 Sep to 11 Oct 2013		
Model	Score	Value	Model	Score	Value	Model	Score	Value
Uninformed	FS_0	-0.70	Uninformed	FS_0	-0.70	Uninformed	FS_0	-2.1
	$\widetilde{FS}$	0		$\widetilde{FS}$	0		$\widetilde{FS}$	0
Alarm-based	$\widetilde{FS}$	-0.04	Alarm-based	$\widetilde{FS}$	0.22	Alarm-based	$\widetilde{FS}$	0.29
Calibrated	$\widetilde{FS}$	0.39	Calibrated	$\widetilde{FS}$	0.34	Calibrated	$\widetilde{FS}$	0.27

30 Sep to 16 Oct 2013			3 to 19 Oct 2013			19 Apr to 5 May 2016		
Model	Score	Value	Model	Score	Value	Model	Score	Value
Uninformed	FS_0	-2.1	Uninformed	FS_0	-1.5	Uninformed	FS_0	-0.70
	$\widetilde{FS}$	0		$\widetilde{FS}$	0		$\widetilde{FS}$	0
Alarm-based	$\widetilde{FS}$	0.28	Alarm-based	$\widetilde{FS}$	0.53	Alarm-based	$\widetilde{FS}$	-0.31
Calibrated	$\widetilde{FS}$	0.27	Calibrated	$\widetilde{FS}$	0.42	Calibrated	$\widetilde{FS}$	-0.14

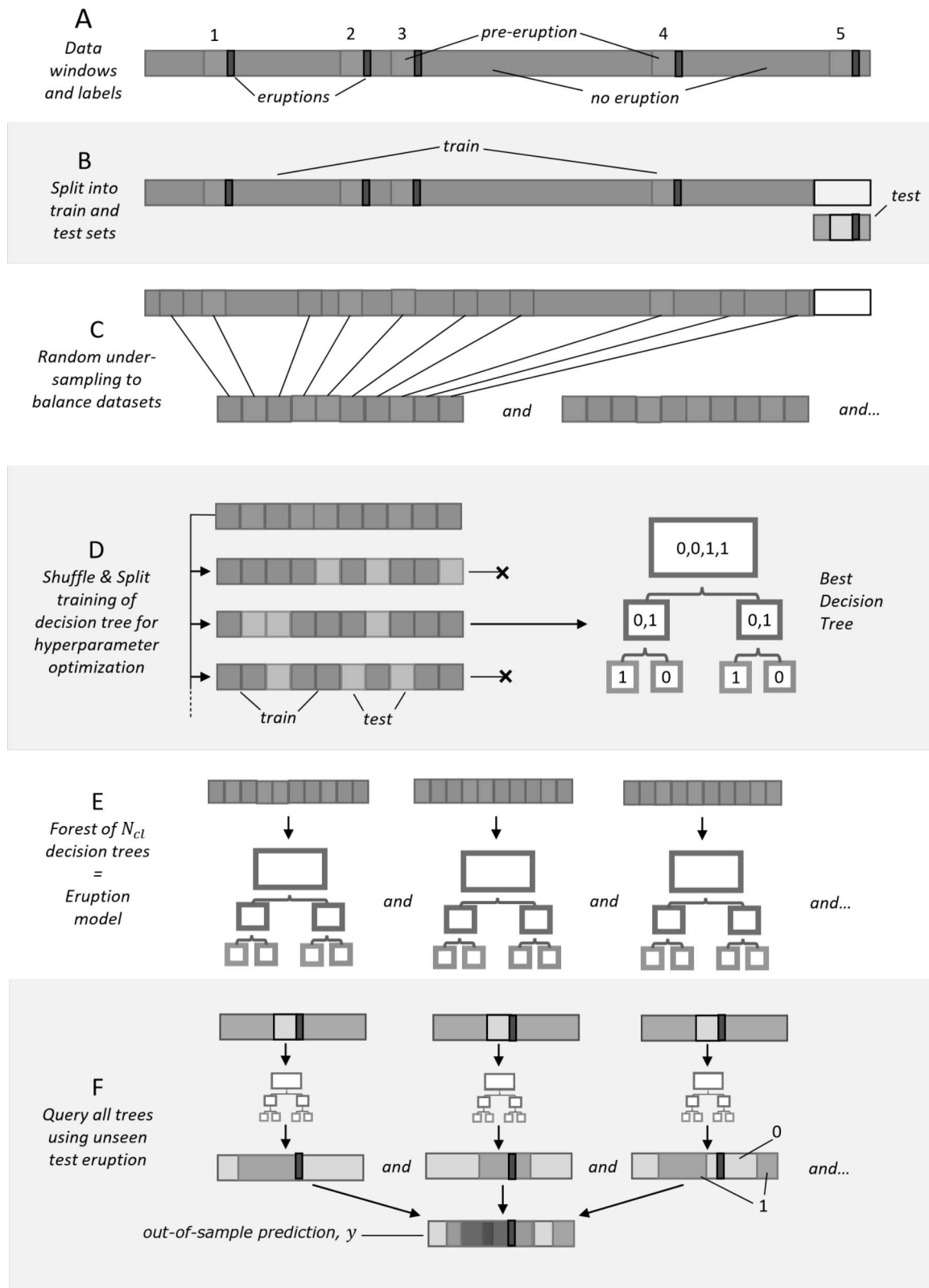


Fig. S1 Model training A. Labeling data windows as eruption (red), pre-eruptive (orange) and non-eruptive. B. Splitting a test eruption (grey/purple) from the training data (orange/green). C. Undersampling the non-eruptive windows to create multiple balanced datasets. D. For each undersampled dataset, use Shuffle-Split to train a decision tree. E. Eruption model as a Random Forest of decision trees specialized on eruptive windows. F. Testing each decision tree on unseen data and averaging their output as the forecast, y . Adapted from Dempsey et al. (2020).

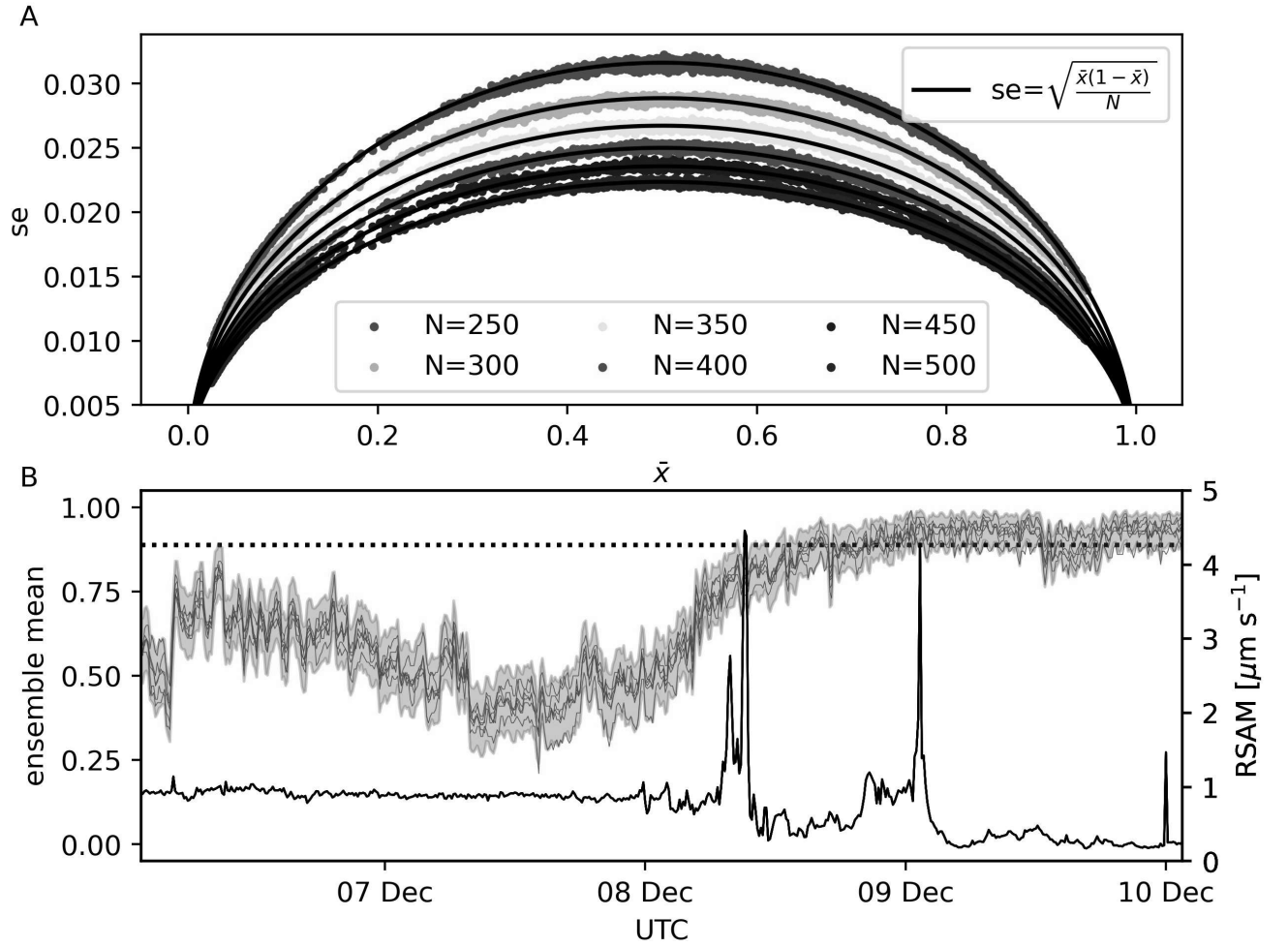


Fig. S2 Comparison of theoretical and numerically constructed confidence intervals. (A) Standard error (se) as a function of estimated proportion ($\bar{x}$) for different N . Colored circles show results of the numerical trials while black lines give the theoretical predictions. (B) A 95% confidence interval for a forecast model with $N = 100$ decision trees (red shading) and five equivalent forecast models constructed with different random number generator seeds (red profiles). RSAM (real-time seismic amplitude measurement) is shown in black.

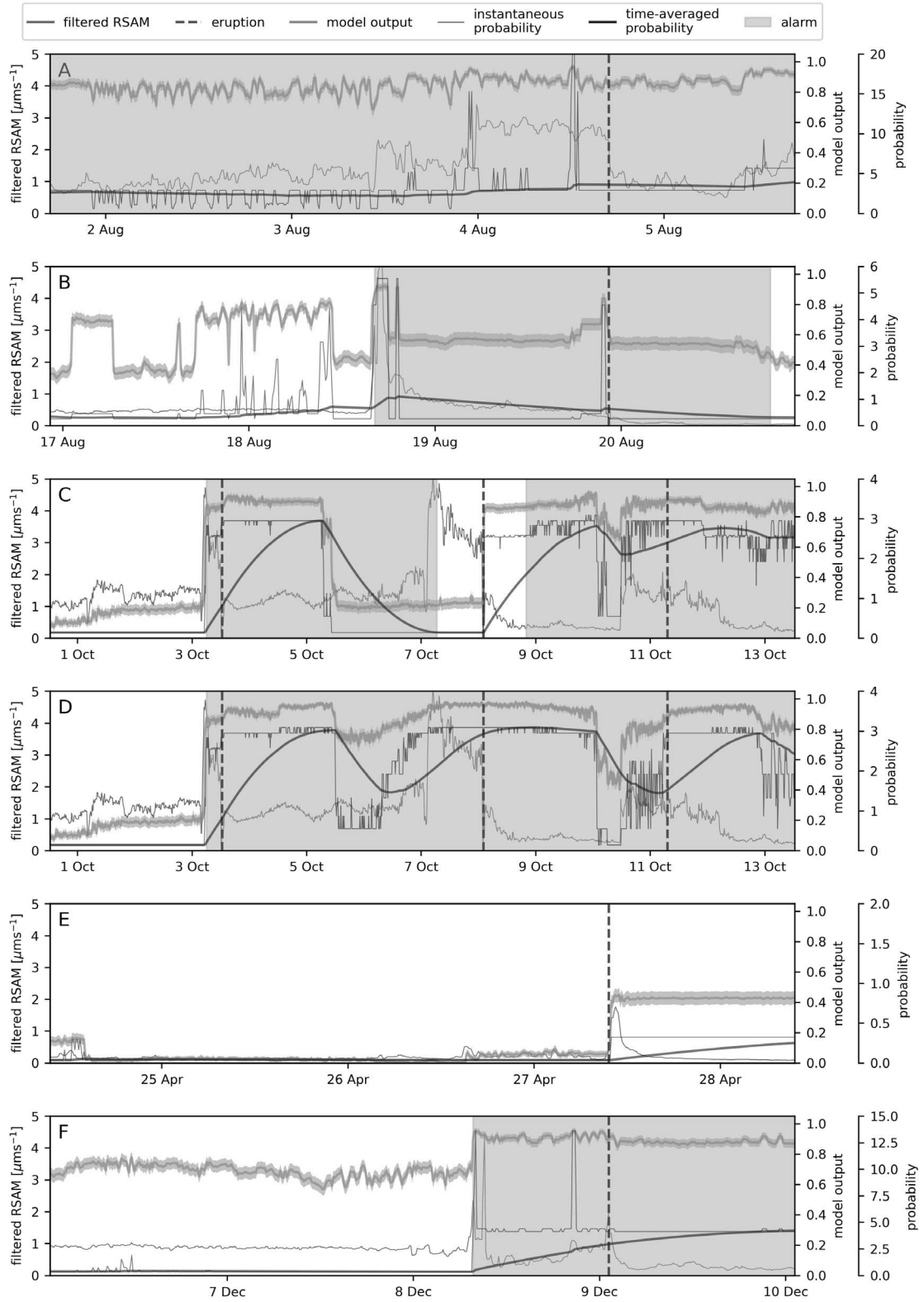


Fig. S3 Out-of-sample performance by the model on five eruptive periods and seven eruptions in (A) 2012, (B) Aug 2013, (C-D) Oct 2013, (E) 2016 and (F) 2019. The effect of retraining a model on the first eruption in a sequence of three is shown in (D). Black = RSAM (real-time seismic amplitude measurement) filtered for earthquakes, cyan = model output with 90% uncertainty envelope, narrow blue = instantaneous probability, thick blue = time-averaged probability, red dashed = eruption, yellow block = alarm period.

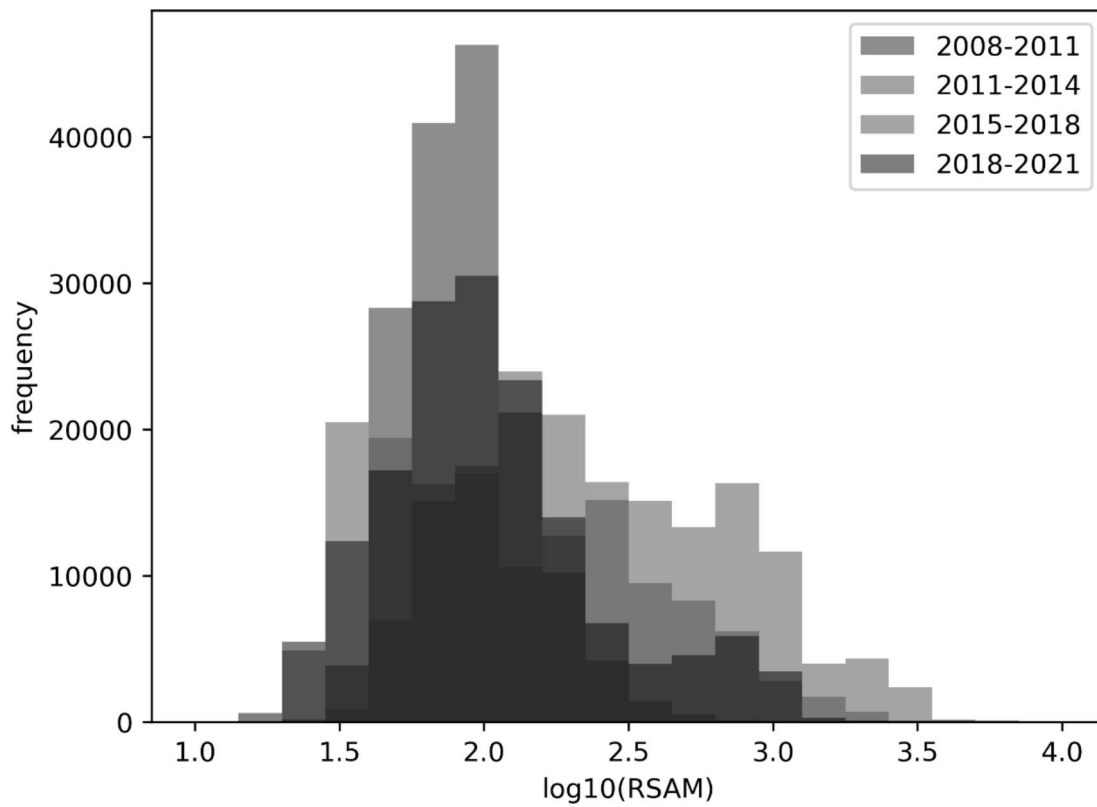


Fig. S4 Distribution of logged RSAM values for four 3-year periods. The periods 2008-2011 (blue) and 2011-2014 (green) summarize data recorded prior to a station upgrade in 2014. The periods 2015-2018 (red) and 2018-2021 (black) summarize data recorded after.